Post-quantum side-channel security from hard physical learning problems

Clément Hoffmann

January 17, 2025





Post-quantum side-channel security from hard physical learning problems



Post-quantum *side-channel security from hard physical learning problems*



Post-quantum side-channel security
from hard physical learning problems



Post-quantum side-channel security from hard physical learning problems

Outline



- **Post-quantum cryptography**
  - Introduction to cryptography
  - Public-key cryptography
  - Post-quantum cryptography

Outline



- **Post-quantum cryptography**
  - Introduction to cryptography
  - Public-key cryptography
  - Post-quantum cryptography
- **Side-channel analysis**
  - Introduction to side-channel information
  - Leakage attacks and countermeasures
  - Side-channel analysis of post-quantum schemes

Outline



- **Post-quantum cryptography**
  - Introduction to cryptography
  - Public-key cryptography
  - Post-quantum cryptography
- **Side-channel analysis**
  - Introduction to side-channel information
  - Leakage attacks and countermeasures
  - Side-channel analysis of post-quantum schemes
- **Learning problems**
  - Definition
  - Leakage attacks and countermeasures
  - Learning problems, side-channel analysis and post-quantum cryptography

Outline



- **Post-quantum cryptography**
  - Introduction to cryptography
  - Public-key cryptography
  - Post-quantum cryptography
- **Side-channel analysis**
  - Introduction to side-channel information
  - Leakage attacks and countermeasures
  - Side-channel analysis of post-quantum schemes
- **Learning problems**
  - Definition
  - Leakage attacks and countermeasures
  - Learning problems, side-channel analysis and post-quantum cryptography

Antique cryptography

First military cryptographic device known



Scytale

Antique cryptography

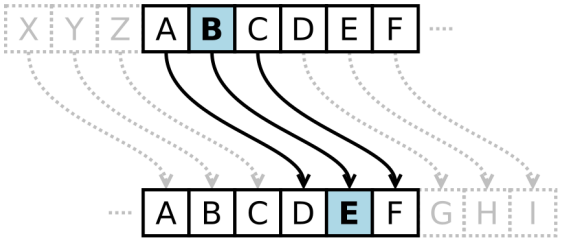
First military cryptographic device known



Scytale

Key = diameter of the stick

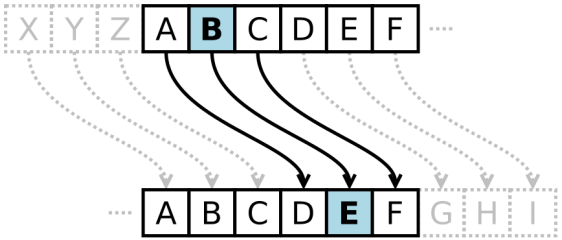
Antique cryptography



Caesar cipher

- Monoalphabetic substitution cipher
- Key = shift parameter

Antique cryptography



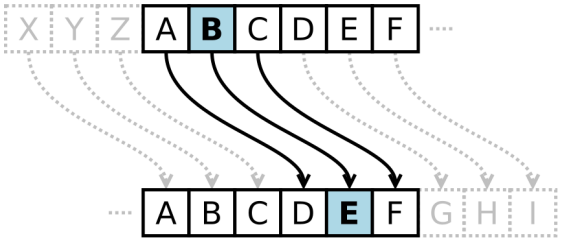
Caesar cipher

CRYPTOGRAM

GVCTXSKVEQ

- Monoalphabetic substitution cipher
- Key = shift parameter

Antique cryptography



Caesar cipher

CRYPTOGRAM
DSZQUPHSBN
ETARVQITCO
FUBSWRJUDP
GVCTXSKVEQ
HWDUYTLWFR
IXEVZUMXGS
...

- Monoalphabetic substitution cipher
- Key = shift parameter

Modern cryptography



German model of an Enigma machine from 1942

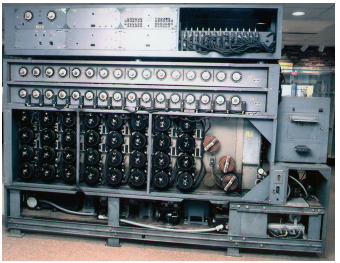
- Settings changing everyday among $\approx 10^{20}$ possibilities
- Permutation composition

Modern cryptography



German model of an Enigma machine from 1942

- Settings changing everyday among $\approx 10^{20}$ possibilities
- Permutation composition



US version of a Turing bomb

Cryptanalysis using computing machine

Kerckhoffs's principle



"A cryptosystem should be secure, even if everything about the system, except the key, *is public knowledge."*

– Auguste Kerckhoffs

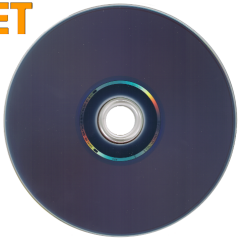
Every day use



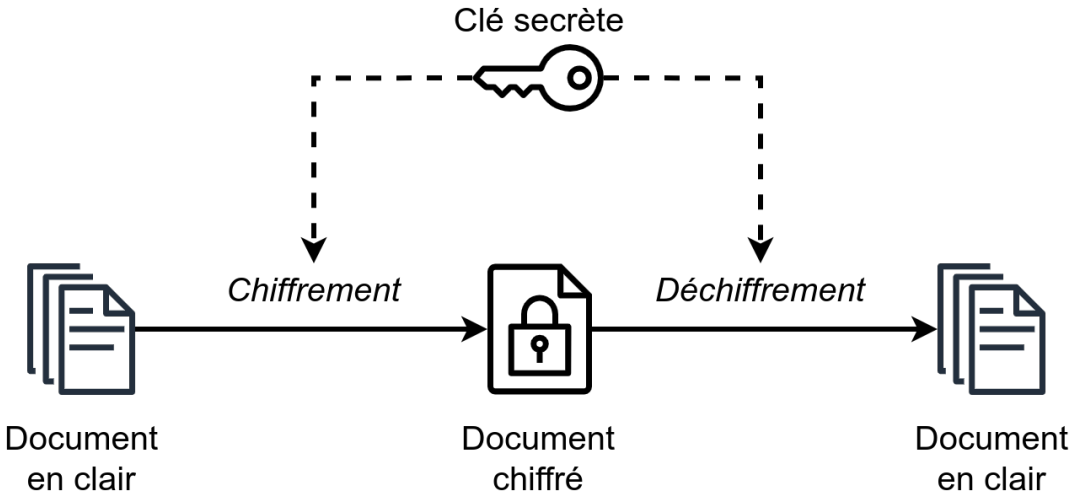




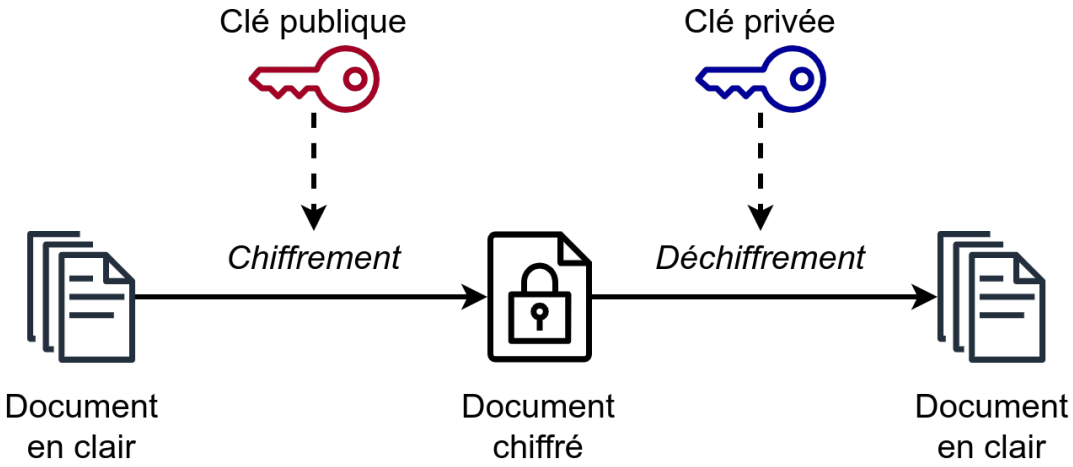
INTERNET



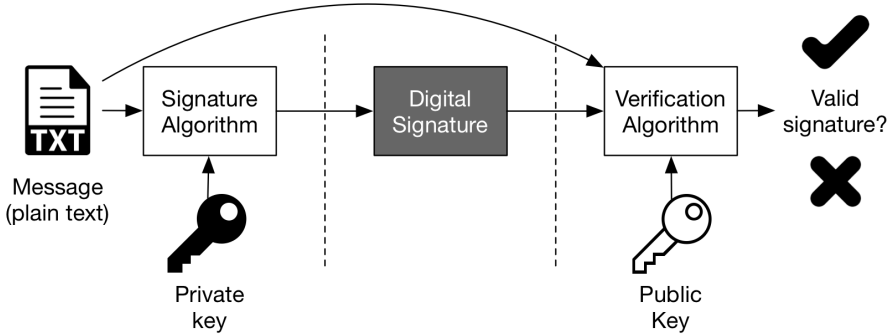
Encryption scheme



Public-key encryption scheme



Signature



- **Authenticity**
- **Integrity**
- **Non-repudiation**



Zero Knowledge Proof

Proving a given statement, without conveying any more information more than the fact that the statement is true.

Examples: password verification, financial transactions...



Zero Knowledge Proof

Proving a given statement, without conveying any more information more than the fact that the statement is true.

Examples: password verification, financial transactions...

Fully Homomorphic Encryption

Performing computations on encrypted data, without decrypting it.

Examples: electronic voting, online privacy...

Quantum computer



Google Quantum AI's quantum computer

Quantum computer



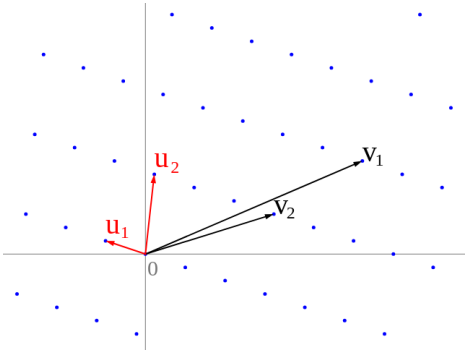
Google Quantum AI's quantum computer

- Exponentially accelerates certain kind of computations
- Breaks most of public-key cryptography

  Opacity in its development

  Harvest now, decrypt later

Post-quantum cryptography



Quantum-resistant problem:

Given a random basis (eg. $\boldsymbol{u}_1, \boldsymbol{u}_2$*),*
it is hard* *to find a basis composed of small norm vectors (eg.* $\boldsymbol{v}_1, \boldsymbol{v}_2$*).*

Post-Quantum Cryptography Standardization



- public, open process for selecting standard for key exchange and signature
- 69 candidates in 2017
- 3 standards in 2024
- creates activity in a field of research, exhort to set parameters and look into concrete propostions

Outline



- **Post-quantum cryptography**
  - Introduction to cryptography
  - Public-key cryptography
  - Post-quantum cryptography
- **Side-channel analysis**
  - Introduction to side-channel information
  - Leakage attacks and countermeasures
  - Side-channel analysis of post-quantum schemes
- **Learning problems**
  - Definition
  - Leakage attacks and countermeasures
  - Learning problems, side-channel analysis and post-quantum cryptography

Side-channel information: the example of Hanabi



Side-channel information: the example of Hanabi



- Context conveys more **information** than actual hint
- **Unintended** information may "leak"

Side-channel information: the example of Hanabi



- Context conveys more **information** than actual hint
- **Unintended** information may "leak"
- *Wait a Second: Playing Hanabi without Giving Hints* - Eger et Gruss

Side-channel information: an adversarial case



Algorithm 1 Check 4-Digit PIN

Require: A 4-digit PIN as an array `PIN[1..4]`, and a function `isValidDigit(digit)` that returns true if the digit is valid.

Ensure: Return `true` if all digits are valid, otherwise `false`.

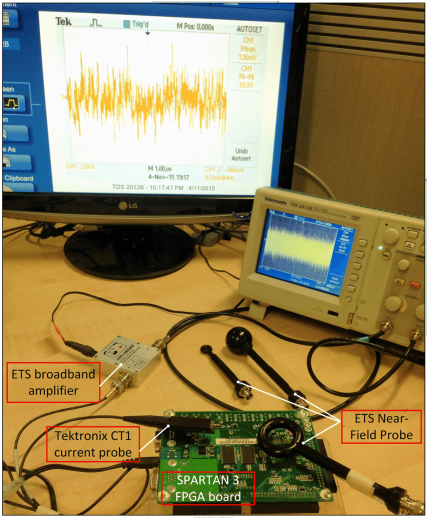
```
1: valid ← true
2: for i ← 1 to 4 do
3:   if NOT isValidDigit(PIN[i]) then
4:     valid ← false
5:     return false
6:   end if
7: end for
8: return valid
```

- Checking all possible PINs:
  10^4 **possibilities**

- By measuring the runtime of the algorithm, we can know how many checks were performed

↪ We can test the digits one by one:
4×10 **possibilities**

Leakage attacks



Leakage attacks: RSA example



Algorithm 2 Square-and-Multiply Algorithm

Require: x, **key** $s = (s_{n-1}, \ldots, s_0)$ in binary
Ensure: x^e

1: $result \leftarrow 1$
2: **for** $i \leftarrow n-1$ **to** 0 **do**
3: $\quad result \leftarrow square(result)$
4: $\quad$ **if** $s_i = 1$ **then**
5: $\quad\quad result \leftarrow multiply(result, x)$
6: $\quad$ **end if**
7: **end for**
8: **return** $result$

Leakage attacks: RSA example

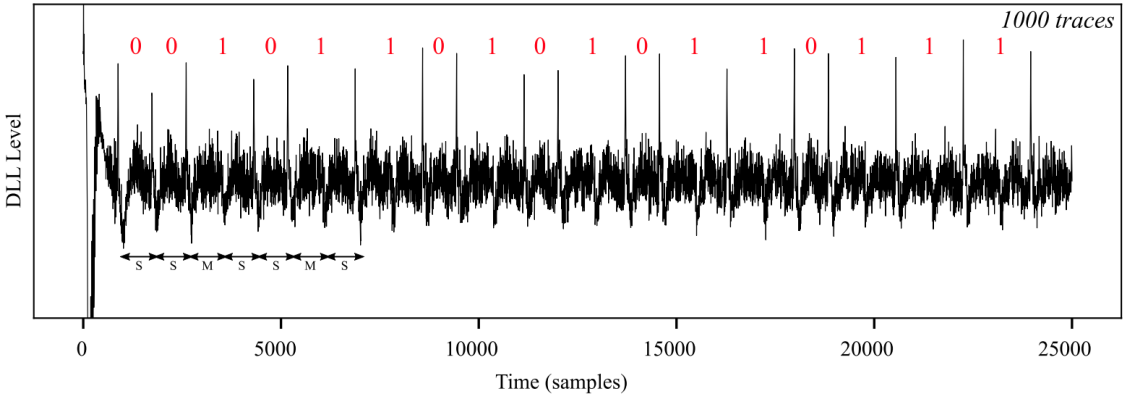


Algorithm 2 Square-and-Multiply Algorithm

Require: x, **key** $s = (s_{n-1}, \ldots, s_0)$ in binary
Ensure: x^e

1: $result \leftarrow 1$
2: **for** $i \leftarrow n-1$ **to** 0 **do**
3: $\quad result \leftarrow square(result)$
4: $\quad$ **if** $s_i = 1$ **then**
5: $\quad\quad result \leftarrow multiply(result, x)$
6: $\quad$ **end if**
7: **end for**
8: **return** $result$

Leakage attacks: RSA example



EM trace on the execution of Square-and-Multiply

Countermeasures



Software:

- Constant-time implementation
- Randomize intermediate values
- Randomize operations order

Countermeasures



Software:

- Constant-time implementation
- Randomize intermediate values
- Randomize operations order

Hardware:

- Noise generation
- Shielding
- High parallelism

Side-channel analysis of post-quantum schemes



My work at the **scheme level** can be found in three papers:

Side-channel analysis of post-quantum schemes



My work at the **scheme level** can be found in three papers:

- We showed that Kyber, NIST (now) standard for *key sharing*, has built-in flaws against side-channel attacks.

Side-channel analysis of post-quantum schemes



My work at the **scheme level** can be found in three papers:

- We showed that Kyber, NIST (now) standard for *key sharing*, has built-in flaws against side-channel attacks.
- We noticed that Dilithium, NIST (now) standard for *signature*, doesn't have the same flaw. We provided the most efficient SCA-secure implementation.

Side-channel analysis of post-quantum schemes



My work at the **scheme level** can be found in three papers:

- We showed that Kyber, NIST (now) standard for *key sharing*, has built-in flaws against side-channel attacks.
- We noticed that Dilithium, NIST (now) standard for *signature*, doesn't have the same flaw. We provided the most efficient SCA-secure implementation.
- We came up with a new scheme implementing ideas to fix Kyber's flaws.

Side-channel analysis of post-quantum schemes



My work at the **scheme level** can be found in three papers:

- We showed that Kyber, NIST (now) standard for *key sharing*, has built-in flaws against side-channel attacks.
- We noticed that Dilithium, NIST (now) standard for *signature*, doesn't have the same flaw. We provided the most efficient SCA-secure implementation.
- We came up with a new scheme implementing ideas to fix Kyber's flaws.

Attack Models in Cryptography



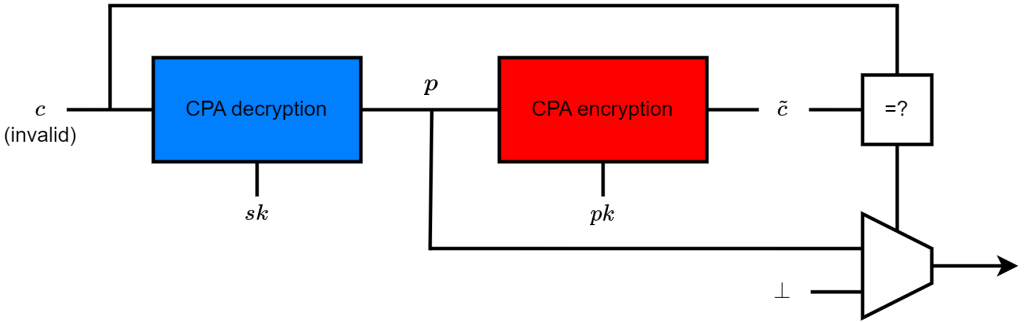
Chosen-Plaintext Attack (CPA)

The attacker can **encrypt** any message.

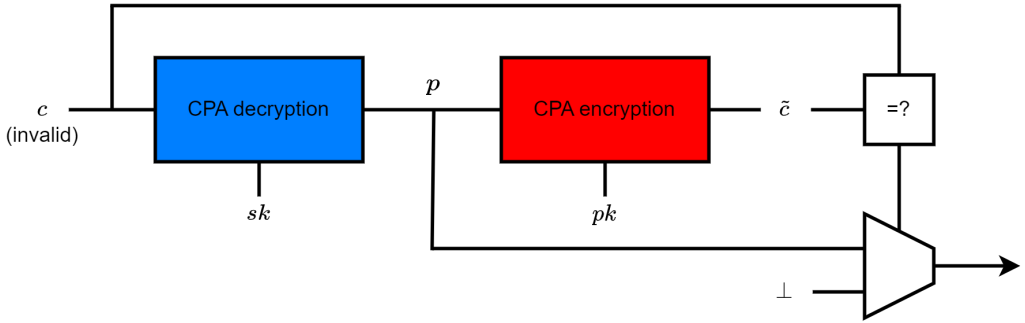
Chosen-Ciphertext Attack (CCA)

The attacker can **encrypt and decrypt** any message or ciphertext.

Kyber - FO-transform

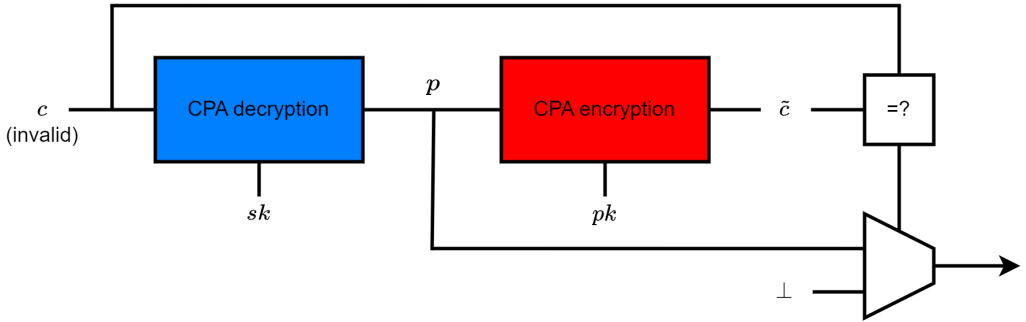


Kyber - FO-transform



Differential Power Analysis (DPA) attack path

Kyber - FO-transform



Differential Power Analysis (DPA) attack path

Distinguishing attack path

Outline



- **Post-quantum cryptography**
  - Introduction to cryptography
  - Public-key cryptography
  - Post-quantum cryptography
- **Side-channel analysis**
  - Introduction to side-channel information
  - Leakage attacks and countermeasures
  - Side-channel analysis of post-quantum schemes
- **Learning problems**
  - Definition
  - Leakage attacks and countermeasures
  - Learning problems, side-channel analysis and post-quantum cryptography

Linear system

$$\begin{cases} 2x + 3y - z = 5 \\ 4x - y + 2z = 6 \\ -x + 5y + 3z = 7 \end{cases}$$

Linear system

In general, a system of m linear equations with n unknowns can be written as:

$$\begin{cases} r_{1,1}x_1 + r_{1,2}x_2 + \cdots + r_{1,n}x_n = b_1 \\ r_{2,1}x_1 + r_{2,2}x_2 + \cdots + r_{2,n}x_n = b_2 \\ \vdots \\ r_{m,1}x_1 + r_{m,2}x_2 + \cdots + r_{m,n}x_n = b_m \end{cases}$$

Where:

$x_1, x_2, \ldots, x_n$: Unknown variables,

$r_{i,j}$: Coefficients of the variables,

b_i: Constant terms.

Linear system

In general, a system of m linear equations with n unknowns can be written as:

$$\begin{cases} r_{1,1}x_1 + r_{1,2}x_2 + \cdots + r_{1,n}x_n = b_1 \\ r_{2,1}x_1 + r_{2,2}x_2 + \cdots + r_{2,n}x_n = b_2 \\ \vdots \\ r_{m,1}x_1 + r_{m,2}x_2 + \cdots + r_{m,n}x_n = b_m \end{cases}$$

Where:

$x_1, x_2, \ldots, x_n$: Unknown variables,

$r_{i,j}$: Coefficients of the variables,

b_i: Constant terms.

Solving a linear system is easy

Learning problems

$$\begin{cases} r_{1,1}x_1 + r_{1,2}x_2 + \cdots + r_{1,n}x_n \approx b_1 \\ r_{2,1}x_1 + r_{2,2}x_2 + \cdots + r_{2,n}x_n \approx b_2 \\ \vdots \\ r_{m,1}x_1 + r_{m,2}x_2 + \cdots + r_{m,n}x_n \approx b_m \end{cases}$$

Where:

$x_1, x_2, \ldots, x_n$: Unknown variables,

$r_{i,j}$: Coefficients of the variables,

b_i: Constant terms.

Learning problems

$$\begin{cases} r_{1,1}x_1 + r_{1,2}x_2 + \cdots + r_{1,n}x_n \approx b_1 \\ r_{2,1}x_1 + r_{2,2}x_2 + \cdots + r_{2,n}x_n \approx b_2 \\ \vdots \\ r_{m,1}x_1 + r_{m,2}x_2 + \cdots + r_{m,n}x_n \approx b_m \end{cases}$$

Where:

$x_1, x_2, \ldots, x_n$: Unknown variables,

$r_{i,j}$: Coefficients of the variables,

b_i: Constant terms.

Solving a linear system with approximate equations is hard

Learning With Errors

$$\begin{cases} r_{1,1}x_1 + r_{1,2}x_2 + \cdots + r_{1,n}x_n = b_1 + e_1 \\ r_{2,1}x_1 + r_{2,2}x_2 + \cdots + r_{2,n}x_n = b_2 + e_2 \\ \vdots \\ r_{m,1}x_1 + r_{m,2}x_2 + \cdots + r_{m,n}x_n = b_m + e_m \end{cases}$$

Where e_i follows a Gaussian distribution.

As hard as lattice problems.

Learning With Rounding

$$\begin{cases} r_{1,1}x_1 + r_{1,2}x_2 + \cdots + r_{1,n}x_n = \lfloor b_1 \rceil \\ r_{2,1}x_1 + r_{2,2}x_2 + \cdots + r_{2,n}x_n = \lfloor b_2 \rceil \\ \vdots \\ r_{m,1}x_1 + r_{m,2}x_2 + \cdots + r_{m,n}x_n = \lfloor b_m \rceil \end{cases}$$

Where $\lfloor \cdot \rceil$ is the rounding function. The approximation is *deterministic*, and can be seen as a loss of information.

As hard as LWE.

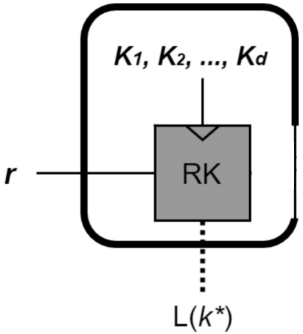
Learning With Physical Rounding (LWPR)



$$\begin{cases} r_{1,1}x_1 + r_{1,2}x_2 + \cdots + r_{1,n}x_n = L(b_1) \\ r_{2,1}x_1 + r_{2,2}x_2 + \cdots + r_{2,n}x_n = L(b_2) \\ \vdots \\ r_{m,1}x_1 + r_{m,2}x_2 + \cdots + r_{m,n}x_n = L(b_m) \end{cases}$$

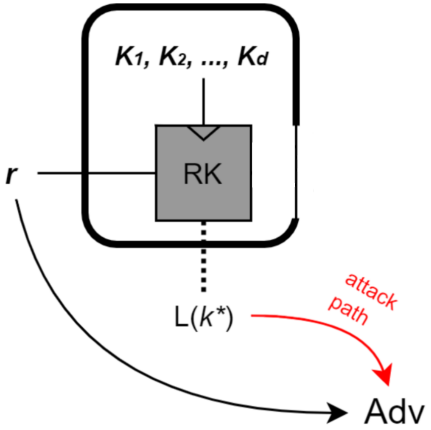
Where L is a leakage function, ie the value we get with an oscilloscope.

Learning With Physical Rounding - Usecase



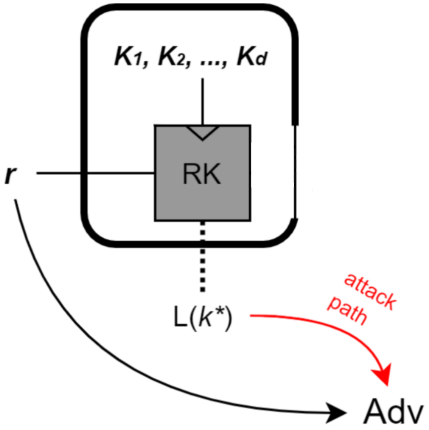
LWPR adversarial model

Learning With Physical Rounding - Usecase



LWPR adversarial model

Learning With Physical Rounding - Usecase

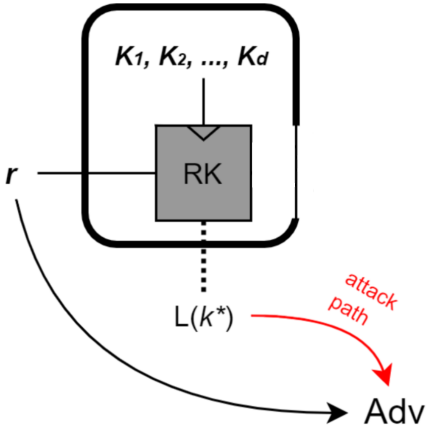


LWPR adversarial model

Useful for:

- Rekeying
- Side-channel analysis of post-quantum scheme

Learning With Physical Rounding - Usecase



LWPR adversarial model

Useful for:

- Rekeying
- Side-channel analysis of post-quantum scheme

How to define:

- Leakage model
- Measurement noise

How to analyze:

- Heuristic hardness
- Asymptotic proves



Thank You!

Figure sources



Caesar - Par Cepheus — Travail personnel, Domaine public, https://commons.wikimedia.org/w/index.php?curid=1235339
Scytale - CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=1698345
Enigma - Nicoleon, CC0, via Wikimedia Commons - https://commons.wikimedia.org/wiki/File:Enigma_1942.jpg
PKE - https://commons.wikimedia.org/w/index.php?curid=128650936
Signature - https://exploitedbunker.com/articles/verifying-digital-signatures/
SCA setup - https://www.researchgate.net/figure/Measurement-setup-for-side-channel-attack_fig4_303097684
RSA SPA - Remote hardware attacks on connected devices - https://theses.hal.science/tel-03879252v1/document